



MTH204 : Symmetry (2024-25)

Tutorial 04 (February 14, 2025)

Be sure that you appreciate the following groups and their subgroups. It will be a good idea to revise them.

- $\text{sym}(\Delta), \text{sym}(\square)$: symmetries of an equilateral triangle and square.
- $\text{rot}(P_n) \leq \text{sym}(P_n)$: symmetries of a regular n -gon and its subgroup of rotational symmetries.
- $\text{rot}(\mathcal{P}) \leq \text{sym}(\mathcal{P})$: symmetries of $\mathcal{P}$ and its subgroup of rotational symmetries, where $\mathcal{P} \in \{\mathbb{T}, C, D, I, O\}$
- $A_n \leq S_n$: permutation group on n letters and its subgroup of even permutations.
- $\text{SO}_3(\mathbb{R}) \leq \text{O}_3(\mathbb{R})$: the group of 3×3 orthogonal matrices with entries in $\mathbb{R}$ and its subgroup of orthogonal matrices with determinant 1.

Recall. Let $H \leq G$. For $g \in G$, the subset $gH := \{gh : h \in H\}$ of G is called the *coset of H in G represented by g* . This subset gH is in bijective correspondence with H . Even when $g_1 \neq g_2 \in G$, it is possible that $g_1H = g_2H$. In fact, $g_1H = g_2H$ if and only if $g_2^{-1}g_1 \in H$. Moreover, if $g_1H \cap g_2H$ is nonempty, then $g_1H = g_2H$. Number of distinct cosets of H in G is denoted by $[G : H]$. It is called the *index of H in G* .

1. Consider the group S_3 .
 - (a). For $H = \{1, (2\ 3)\}$ and $g = (1\ 2\ 3)$, compute gH . How many distinct cosets does H have in G ?
 - (b). For $H = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$ and $g = (1\ 2)$, compute gH . How many distinct cosets does H have in G ?
 - (c). How many subsets of G are cosets for a some subgroup H of G ?
2. Given $H \leq G$ and $g \in G$, define $Hg := \{hg : h \in H\}$. Find an instance when $gH = Hg$, and find an instance when $gH \neq Hg$.
3. Suppose $[G : H] = 2$. Show that $gH = Hg$ for every $g \in G$.
4. **(Lagrange's Theorem)**. Let $H \leq G$. If G is finite, show that $|H|$ divides $|G|$. Further, convince yourself that $[G : H] = |G|/|H|$.
5. Consider $\text{rot}(\mathbb{T})$, the group of rotational symmetries of a tetrahedron. Identify a normal subgroup and describe its cosets.
6. What is the size of the group A_4 ? Is there a subgroup of A_4 containing 4 elements? What about a subgroup containing 6 elements?

7. *True or false?* “For every $H \leq G$ and $g_1, g_2 \in G$, we have $g_2g_1H = g_1g_2H$.”

8. For two cosets g_1H and g_2H , your teacher defines $*$ as follows:

$$g_1H * g_2H := g_1g_2H$$

A student remarks that the definition does not make sense. Student is actually right!
Argue for this student.
